Highlights

Model-Free PID Tuning via Encirclements of the Step Response

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- A deterministic reading of the step response replaces manual PID tuning.
- Three phase portraits of the control error give a dimensionless damping diagnostic.
- The diagnostic is band-selective, assigning oscillation to the I, P, or D channel.
- A well-posedness guard keeps the encirclement count independent of window length.
- One triangular rule converges to the boundary of the well-damped tuning set.

Model-Free PID Tuning via Encirclements of the Step Response

Daniel Pachner[a], Pavel Otta[a,*], Jiří Dostál[a], Vladimír Havlena[b]

[a] *University Centre for Energy Efficient Buildings (UCEEB), Czech Technical University in Prague, Buštěhrad, Czech Republic*
[b] *Department of Control Engineering, Faculty of Electrical Engineering, Czech Technical University in Prague, Prague, Czech Republic*

Abstract

From one closed-loop step response, three phase portraits of the control error are formed—error against its running integral, first difference, and second difference—and the winding number of each about its settling point is counted. The counts are invariant to time and amplitude scaling, so fixed dimensionless thresholds serve every loop, and they are band-selective: they locate insufficient damping in the low-, mid-, and high-frequency bands owned by the integral, proportional, and derivative channels. A settling-anchored window makes the counts independent of the observation length; a parameter-free stability screen halves all gains when the record fails to settle, so the method bootstraps from arbitrary initial gains. A triangular rule cuts the lowest violated band—the only one whose blame is unambiguous—raises the verified-clean bands below it, and with all portraits clean raises all bands. The iteration is deterministic, bounded by construction, and exploration-free—safe on a process in operation—and it converges to a limit cycle on the boundary of the well-damped tuning set; the required monotonicity is derived from the plant class near the boundary. The method rests on two assumptions—a stable, self-regulating process of the first-order-plus-time-delay class and a controller from the nested family I ⊂ PI ⊂ PID with mandatory integral action—and is validated on a battery of process-typical plants with identical defaults throughout; each iteration costs one routine setpoint step, and every decision is visible in the three portraits. A reference implementation is public.

Keywords: PID control, controller tuning, data-driven control, phase plane, step response

1. Introduction

Eighty years after Ziegler and Nichols [1], PID remains the dominant industrial controller [2], and most PID loops are still tuned by inspection: an engineer steps the setpoint, looks at the response, and adjusts a gain by feel. The literature's two remedies both discard the inspection. Model-based rules—Ziegler–Nichols [1], Cohen–Coon [3], IMC [4], SIMC [6]—replace it with an identified low-order model and inherit that model's errors. Both experimental legs of Ziegler–Nichols have since been revisited with modern robustness tools: Åström and Hägglund [7] rederived the step-response method from the standpoint of robust loop shaping, obtaining simple tuning rules with guaranteed robustness for essentially monotone processes, while Schlegel and co-workers [8, 9] revisited the frequency-response method, developing robust PID design over sets of plants that share a structural constraint—for example, a single common point of the frequency response. Data-driven schemes replace it with a criterion and the machinery to serve it: a dedicated relay experiment that deliberately excites a sustained oscillation [5], gradient experiments on a quadratic cost [10], a one-shot identification-like problem [11], or online perturbation of the gains [12]. The inspection itself—which gain is to blame, given this response—has remained tacit.

This paper formalizes the inspection. The contributions are: (i) a triple of phase portraits of the control error whose encirclement counts form a dimensionless, scale-free, band-selective damping diagnostic (Section 2); (ii) a well-posedness analysis of the counts, including a counterexample showing that a naïvely windowed count diverges, and a settling-anchored guard that provably restores window independence (Section 3); (iii) a single triangular tuning rule in band-ordered gain coordinates, interpreted as band-wise loop shaping, with boundedness unconditional and boundary convergence under a monotonicity property derived from the plant class (Proposition 6, Section 4); (iv) validation on a battery of process-typical plants with

*Corresponding author
Email address: `pavel.otta@cvut.cz` (Pavel Otta)

identical default parameters throughout (Section 5).

The method claims no optimality: it returns a fast, well-damped tuning—the contract an expert tuner offers—in exchange for one closed-loop setpoint step per iteration and nothing else. No model, no relay experiment, no solver; and full auditability, since an operator watching the three plots sees exactly what the algorithm sees. The whole method carries five dimensionless constants (Table 2), none adjusted for any plant.

Every move either repairs a diagnosed fault or tightens all gains by one verified notch—no iterate is an exploration step—so the tuner runs safely on a process in operation. Adaptive and learning-based tuners must operate the plant under unvetted or perturbed controllers to acquire information [10, 12]; here information is a by-product of a routine setpoint step under the current controller. Figures 1–4 show the method on one plant (P2, Section 5), from a severely mis-tuned start, before any mathematics: response and portraits before tuning and after, captured from the running implementation (Data availability).

2. The encirclement features

The method rests on exactly two standing assumptions, stated once and used throughout.

Assumption 1 (Process class)**.** *The process is a stable, self-regulating SISO plant whose open-loop step response is essentially monotone with dead time—the class adequately captured by first-order-plus-time-delay (FOPTD) models. FOPTD is meant here as a modelling class, not a literal transfer-function restriction: higher-order overdamped dynamics belong to it.*

"Essentially monotone" admits mild non-monotonicity; the construction extends to plants with damped open-loop oscillations provided a well-damped tuning exists within the gain box and the monotonicity of Proposition 6 holds there. What excludes a plant is a resonance strong enough that reducing a band's gain fails to damp that band. The propositions below are proved on the strictly monotone core of the class; a checkable characterization of the wider admissible set, in the spirit of [8, 9], is left open.

The proofs below use the frequency-domain content of the FOPTD core of Assumption 1: both $|G(j\omega)|$ and $\arg G(j\omega)$ strictly decrease on $(0, \infty)$, with $\arg G \to -\infty$ (dead time); for the canonical member $|G| = K/\sqrt{1 + \omega^2 T^2}$, $\arg G = -\arctan \omega T - \omega L$. Monotone *phase* makes the loss of closed-loop damping a single-frequency event (Proposition 2); monotone *gain* orders the bands the controller channels own (Proposition 4). One structural fact joins them: C is linear in its gains, so scaling all three by m maps $L = CG$ to mL—the Nyquist locus changes magnitude, not shape.

Assumption 2 (Controller structure)**.** *The controller is drawn from the nested family* I ⊂ PI ⊂ PID*: a parallel structure $u = K_p e + K_i \int e + K_d \dot{e}$ in a unity-feedback loop, with the integral channel mandatory ($K_i > 0$) and the proportional and derivative channels optional ($K_p, K_d \geq 0$).*

Nothing else is assumed: no model, no noise model, no excitation beyond routine setpoint steps under the existing controller. The mandatory integral channel is what drives $e \to 0$ at steady state and thus lets the portraits below terminate at the origin; its necessity is made precise after Definition 1.

A tuning experiment applies a setpoint step and records the error e_k, $k = 0, \ldots, K$, at period T_s, over a window in which the response settles. Form the running integral E_k and the differences Δe_k, $\Delta^2 e_k$, and pair each signal with its own difference:

$$\begin{aligned} &\Gamma_0 : (e_k,\ E_k - E_K), \\ &\Gamma_1 : (\Delta e_k,\ e_k), \\ &\Gamma_2 : (\Delta^2 e_k,\ \Delta e_k). \end{aligned} \tag{1}$$

Γ_1 is the classical phase plane of the error; Γ_0 and Γ_2 are its integrated and differentiated siblings; all three terminate at the origin when the loop settles. We call (1) *Pachner plots*, after the first author, who devised the construction.

A well-damped loop spirals into the origin without completing a turn; an under-damped loop encircles it. The count is made precise as follows.

Definition 1 (Encirclement count)**.** *Given a planar trajectory (x_k, y_k), $k = 0, \ldots, J$: (i) normalize each coordinate by its peak magnitude, mapping the curve into $[-1, 1]^2$; (ii) truncate at the last entry into the disc of radius $\varepsilon \in (0, 1)$ about the origin; (iii) define N as the winding number of the remaining curve about the origin: total swept angle over 2π, evaluated as the endpoint angle difference corrected by signed crossings of the negative horizontal semi-axis. Partial revolutions count fractionally.*

Definition 1 presupposes that the trajectory terminates at the origin, which the integral channel guarantees: $e \to 0$ for every stable tuning of Assumption 2. Without it, a P-only loop on a self-regulating plant settles at a nonzero offset, Γ_0 and Γ_1 terminate away from

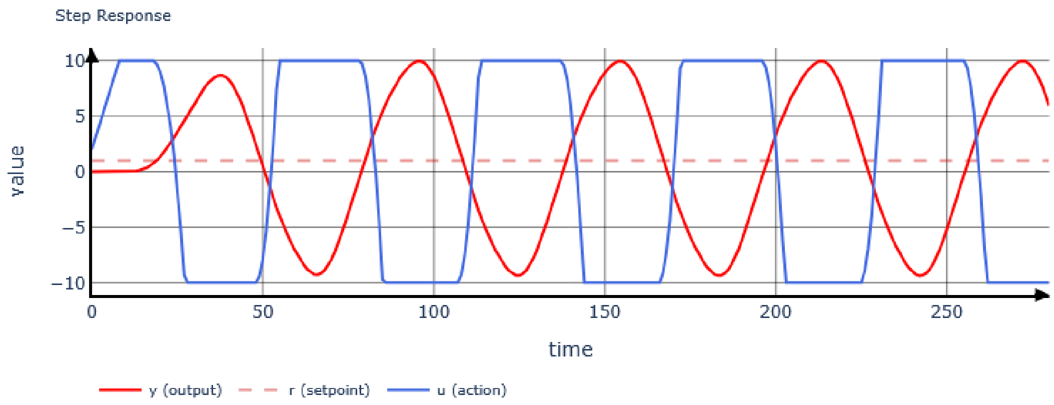

Figure 1: Step response before tuning. The mis-tuned derivative channel sustains an oscillation with the actuator railing between its limits; nothing here suggests which gain is at fault.

the origin, truncation never triggers, and the winding number measures the wrong quantity. This is the concrete content of requiring $K_i > 0$.

Applying Definition 1 to $\Gamma_0, \Gamma_1, \Gamma_2$ yields the feature vector $\mathbf{N}(\theta) = (N_0, N_1, N_2)$ at controller setting $\theta = (K_p, K_i, K_d)$, from a single step test, with no plant knowledge.

Proposition 1 (Invariance)**.** *$\mathbf{N}$ is invariant under time reparametrization $t \mapsto \alpha t$, $\alpha > 0$, of the underlying response, and under independent positive scaling of either coordinate of each trajectory—in particular under scaling of the setpoint step and, in the fast-sampling limit, of the sampling period.*

Proof. The winding number depends only on the oriented image of the curve, not its parametrization; per-axis peak normalization removes amplitude and units, and the truncation disc is defined on normalized coordinates. □

Consequently one threshold set serves all loops: the defaults $\bar{\mathbf{N}} = (0.5, 0.75, 1.0)$ together with $\varepsilon = 0.1$ are used unchanged in every result of this paper. They function as the definition of "well damped," not as tuning knobs.

Which damping the counts read, and why a single one dominates, is a consequence of the plant class.

Proposition 2 (Critical pair)**.** *Let G satisfy Assumption 1 and scale all gains together, $L \mapsto mL$, $m > 0$. The loop is stable up to a finite m^* and loses stability there through one complex pole pair, at the phase-crossover frequency ω_{180}—the maximizer of $|L(j\omega)|$ over $\{\omega : \arg L(j\omega) = -\pi\}$. Near m^* this pair is the least damped in the loop, and its damping $\zeta(m)$ decreases continuously to zero as $m \uparrow m^*$.*

Proof. $\arg C \in [-\pi/2, \pi/2]$ while $\arg G \to -\infty$, so $\arg L$ reaches $-\pi$ at frequencies bounded away from 0 and ∞. Growing m scales the locus mL radially; it first touches -1 at $m^* = 1/|L(j\omega_{180})|$, generically at one frequency. The touch is transversal ($\partial|mL|/\partial m = |L| > 0$), so one conjugate pair crosses the imaginary axis at ω_{180} while all other poles keep strictly positive damping nearby; continuity in m gives the strict decrease of $\zeta(m)$. □

So near the boundary one pair carries the ringing; write

$$e(t) = A\,e^{-\alpha t}\cos(\omega t + \varphi) + r(t), \qquad \zeta = \frac{\alpha}{\sqrt{\alpha^2 + \omega^2}}, \tag{2}$$

with r the remaining, faster-settling part of the error. The decomposition is exact as $m \uparrow m^*$; between the well-damped boundary and the stability boundary—where the rule operates—it is the working model, validated in Section 5. The propositions below read (2).

Proposition 3 (Damping law)**.** *For the mode of* (2)*, each truncated portrait of Definition 1 winds*

$$N = \frac{\ln(1/\varepsilon)}{2\pi}\,\frac{\sqrt{1-\zeta^2}}{\zeta} + \eta, \qquad |\eta| < 1, \tag{3}$$

where η is a fractional endpoint term. The map $\zeta \mapsto N$ is strictly decreasing on $(0, 1)$; each threshold crossing $N_k = \bar{N}_k$ therefore occurs at a unique damping $\bar{\zeta}_k$.

Proof. Differentiation multiplies the mode by the fixed complex number $-\alpha + j\omega$, so each portrait is a fixed linear image of the mode's phase plane; a linear isomorphism preserves oriented loops, and all three portraits

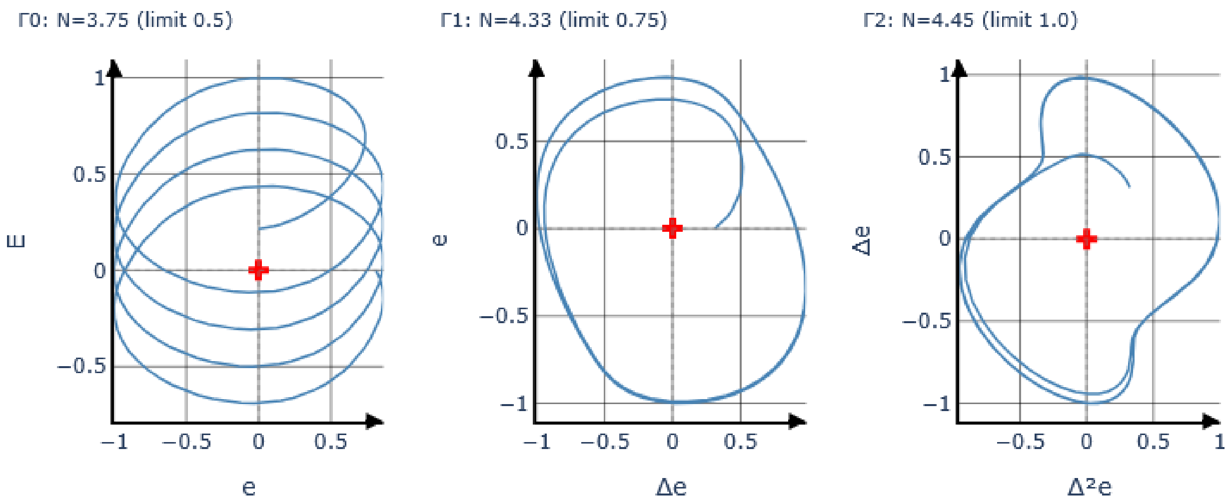

Figure 2: Portraits before tuning, left to right $\Gamma_0, \Gamma_1, \Gamma_2$. All three loop well past their limits (N_0=3.75 vs. 0.5; N_1=4.33 vs. 0.75; N_2=4.45 vs. 1.0)—the response of Figure 1 is unreadable, but here the fault is explicit in every band at once.

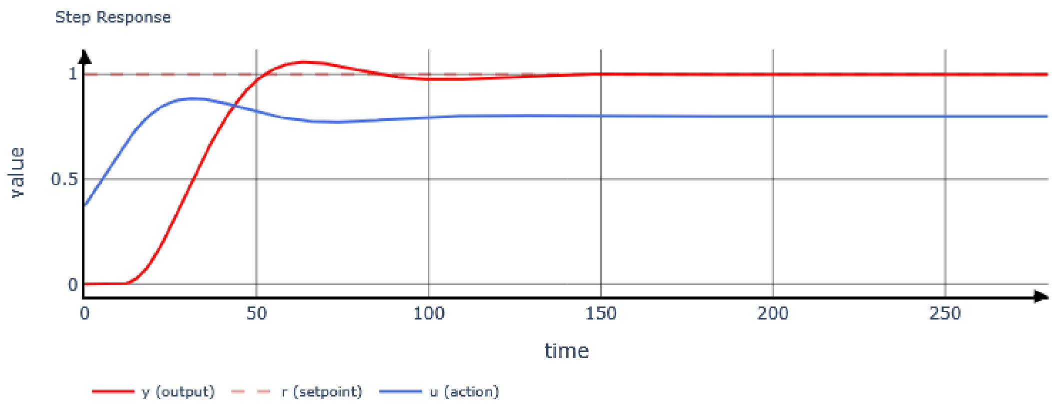

Figure 3: Step response after tuning: a single overshoot settling cleanly to the setpoint, actuator unsaturated.

trace one logarithmic spiral up to linear equivalence. From its peak to the last entry into the ε-disc the spiral's radius contracts by ε, taking time $\alpha^{-1}\ln(1/\varepsilon)$, i.e. $(\ln(1/\varepsilon)/2\pi)(\omega/\alpha)$ revolutions, and $\omega/\alpha = \sqrt{1-\zeta^2}/\zeta$; the truncated endpoints contribute less than a half-turn each. Monotonicity: $\mathrm{d}(\sqrt{1-\zeta^2}/\zeta)/\mathrm{d}\zeta < 0$ on $(0,1)$. $\square$

Hence $N_k > \bar{N}_k \iff \zeta < \bar{\zeta}_k$: "rings" is a sign test, scale- and clock-free by Proposition 1. The rule uses only the sign; inverting (3) gives a free damping estimator. Sampling shifts the constant in (3)—most in Γ_2—but never the monotonicity, so the damping specifications are *calibrated*, not derived: the measured $N_k(\zeta)$ of the exact second-order step error crosses $\bar{\mathbf{N}} = (0.5, 0.75, 1.0)$ at $\bar{\zeta} \approx 0.64, 0.49, 0.24$ ($\varepsilon = 0.1$), strictest in the lowest band. For a clean dominant mode the count is set by ε; the settling window of Section 3 only guards against noise-dominated tails.

Proposition 4 (Band selectivity and upward leakage)**.** *Write $e = x + r$ as in (2). Each differencing step multiplies the mode by its natural frequency $\omega_n = \sqrt{\alpha^2+\omega^2}$ and the essentially monotone residual only by its slower decay rate $\alpha_r < \omega_n$; each integration does the reverse. A mode's visibility in Γ_j thus grows by $\omega_n/\alpha_r > 1$ per band index: it is most visible in its own band, visible in every band above, and invisible below. The set of bands whose count a mode lifts is upward-closed, $\{k, k+1, \dots\}$, and several modes superpose, each upward-closed.*

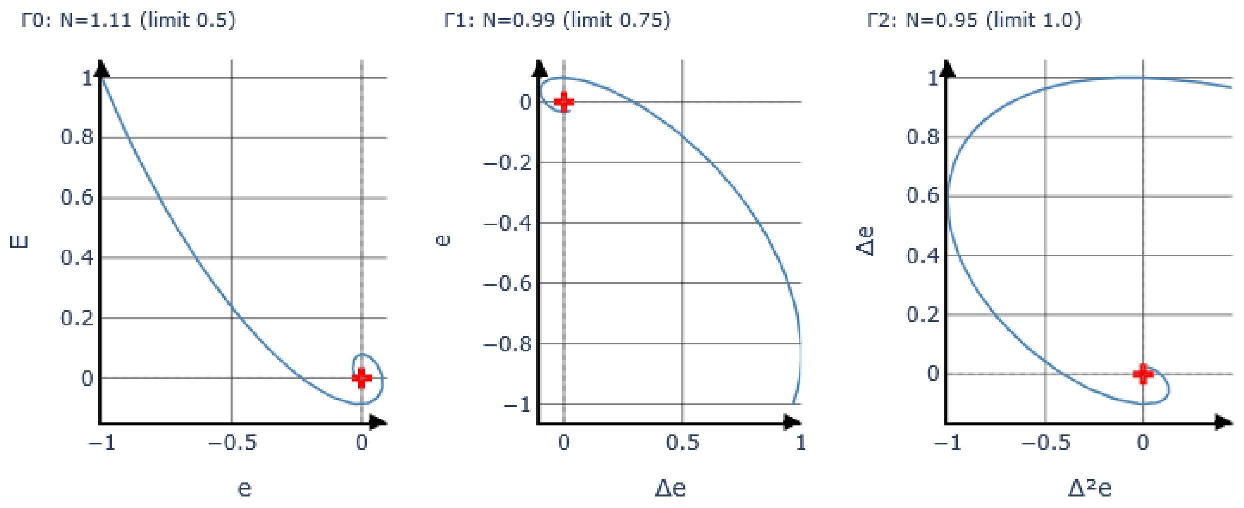

Figure 4: Portraits after tuning. Γ_2 is now within its limit (0.95 vs. 1.0); Γ_0, Γ_1 have fallen sharply from Figure 2 ($3.75 \to 1.11$, $4.33 \to 0.99$) and sit just outside theirs, the boundary limit cycle of Proposition 7 caught mid-oscillation rather than a rare failure to converge.

The bands are owned by the controller channels: for the parallel PID, $|C(j\omega)| \approx K_i/\omega$ at low frequency, $\approx K_p$ in the mid band, $\approx K_d\omega$ at high frequency. Hence N_0 indicts K_i, N_1 indicts K_p, N_2 indicts K_d. This correspondence is the entire basis of the tuning rule.

Attribution. The violated set is a union of upward-closed sets (Proposition 4), so its lowest member is the unique band whose violation cannot be leakage—nothing exists below it to leak in—while violations above it are evidentially void: a genuine upper-band mode and upward leakage produce indistinguishable counts. Synthetically, one under-damped mid mode alone gives $\mathbf{N} = (0.12, 3.9, 6.5)$—the high count exceeds the mid one purely by leakage—while a lone high mode gives $(0.12, 0.05, 5.7)$, leaking nowhere below. The rule of Section 4 is built on this asymmetry.

3. Well-posedness of the count

Per-axis normalization buys Proposition 1 but blinds each portrait to absolute scale. A persistent micro-oscillation—quantization, a marginally damped parasitic mode, numerical residue at a fraction of a percent of the step—can dominate Δe, $\Delta^2 e$ long after e has settled; normalized to its own peak, the ripple fills the portrait and winds indefinitely, so the count grows with the observation window and threshold comparisons become artefacts of the window.

This is not hypothetical. For the plant $P(s) = e^{-4s}/(8s+1)^6$ at gains reached mid-iteration by the tuning rule, the response settles to $|e| < 0.005$ (peak 1.0) well inside the default window, yet the counts evaluate to $\mathbf{N} = (0.12, 1.84, 2.86)$ on that window, $(0.62, 9.22, 17.2)$ on a window six times longer, and $(0.62, 17.3, 37.3)$ on twice that: N_1, N_2 grow essentially linearly in window length. In closed loop the consequence is erratic: threshold comparisons flip under 10% gain changes and the corrective moves of the tuning rule fight each other without converging. Figure 5 shows the pattern: unguarded counts climb without bound as the window lengthens, while the guarded counts defined next sit flat.

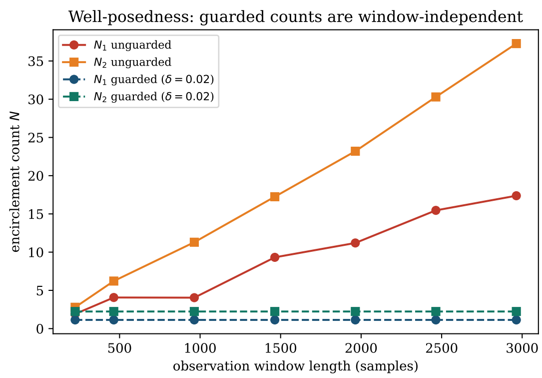

Figure 5: Window dependence of N_1, N_2 on the counterexample plant, guarded (Definition 2) versus unguarded (Definition 1 applied to the raw window). Guarded counts are exactly flat.

The remedy is to anchor the window to the settling of the parent signal.

Definition 2 (Settling-anchored window). *Fix $\delta \in (0,1)$ (default $\delta = 0.02$). Let $k_\delta = 1 + \max\{k : |e_k| > \delta \max_j |e_j|\}$ and truncate all three trajectories* (1) *at k_δ before applying Definition 1.*

Proposition 5 (Well-posedness). *Under Definition 2, the features* **N** *are independent of the window length for every window containing the δ-settling time of e, and Proposition 1 continues to hold (δ is dimensionless and defined on the normalized error).*

Proof. All samples beyond k_δ are discarded, and k_δ itself depends only on e up to its δ-settling time; enlarging the window changes neither. Invariance is preserved because k_δ is defined through the ratio $|e_k| / \max_j |e_j|$. □

On the counterexample above, the guarded counts evaluate to $\mathbf{N} = (0.12, 1.14, 2.24)$ on all three windows—identical to three digits—and, as Section 5 shows, the closed-loop tuning iteration on the same plant converges cleanly once the guard is in place. The guard is henceforth part of the feature definition; the method thus carries exactly three dimensionless constants $(\bar{\mathbf{N}}, \varepsilon, \delta)$, none of which was adjusted for any plant in this paper.

Remark 1. *Proposition 5 formalizes, and enforces, the practical requirement that each tuning experiment be observed until the loop settles. The failure mode it excludes—unbounded winding of a noise-dominated differenced portrait—is silent and window-dependent, which is precisely why it deserves a formal guard rather than an operator's discretion.*

Remark 2 (Scope of the guard). *Proposition 5 assumes a noise-free e. With noise level σ_n, k_δ is well defined only if $\delta \max_j |e_j| \gg \sigma_n$; otherwise k_δ tracks the last noise excursion and the divergence can reappear despite the guard—so $\delta = 0.02$ presumes noise below roughly two percent of the step, a condition to be checked, not assumed. Likewise, retain enough samples per transient (tens to a few hundred) that Δe, $\Delta^2 e$ are not quantization-dominated, and difference the record only from the step instant onward, so that the reference jump injects no kick into Γ_2.*

A stability screen. The guard, and the counts themselves, presuppose that the record settles at all: the portraits of a diverging error never enter the truncation disc, and their winding is an artefact of wherever the recording happened to stop. Whether the record settles is itself decidable from the record, with no model and no thresholds.

Definition 3 (Stability screen). *Model the recorded error as emitted by two zero-mean Gaussian generators switched once at an unknown sample τ. The maximum-likelihood switch point is*

$$\tau^* = \arg\max_\tau \left[-\tau \ln s_1^2(\tau) - (K - \tau) \ln s_2^2(\tau)\right], \tag{4}$$

where $s_1^2(\tau)$ and $s_2^2(\tau)$ are the mean squares of $e_0, \ldots, e_{\tau-1}$ and $e_\tau, \ldots, e_K$ respectively (with τ ranging over splits that leave at least 10% of the record on each side). The record is declared unstable *when $s_2^2(\tau^*) \geq s_1^2(\tau^*)$—the mean-square amplitude of the segment attributed to the second generator is at least that of the first—and* stable *otherwise.*

A settling response front-loads its energy; a diverging one back-loads it. The verdict is invariant to amplitude and time scaling. Zero-mean moments, rather than variances about segment means, handle dead time correctly: the delay plateau is quiet about its own mean but loud about zero, hence read as part of the transient rather than as a silent first generator; the zero reference itself is licensed by Assumption 2, since $K_i > 0$ forces every stable record to end quiet about zero specifically. A bounded sustained oscillation deliberately *passes* the screen: a limit cycle has a frequency, is attributable, and belongs to the band rules, while the screen is reserved for divergence, which no count can describe.

4. The tuning rule

Order the gains by the frequency band each channel dominates and write

$$K^{(0)} = K_i, \qquad K^{(1)} = K_p, \qquad K^{(2)} = K_d, \tag{5}$$

so that the band index aligns with the portraits: N_k is the feature that indicts $K^{(k)}$. Parametrize the gains as multipliers $F^{(k)} \in [F_{\min}, F_{\max}]$ (defaults $[0.01, 10]$) on any conservative stabilizing initialization, fix a step β (default 0.1), and let $\gamma = 1/(1-\beta)$. Each iteration performs one step test, evaluates **N**, and updates the gains.

The rule, in words: *if the record grows rather than decays, halve all three gains; otherwise scan the portraits from slow to fast; the first one that loops too much names the gain to turn down one notch, and every gain below it turns up one notch; if none loops, turn all three up.* Table 1 is the entire decision logic; the commissioning craft is recognizable row by row, and the lower-triangular stencil that names the rule is visible in the arrow pattern.

Table 1: The tuning rule at a glance. The first matching row applies, top to bottom, once per step test; "unstable" is the verdict of Definition 3, "loops" means $N_k > \bar{N}_k$. $\uparrow$: multiply by γ; $\downarrow$: divide by γ; $\Downarrow$: divide by 2; $\cdot$: unchanged. Projections onto $[F_{\min}, F_{\max}]$ implied.

Condition	F_i	F_p	F_d	craft reading
unstable	$\Downarrow$	$\Downarrow$	$\Downarrow$	runaway: halve all
Γ_0 loops	$\downarrow$	$\cdot$	$\cdot$	slow cycling: less reset
Γ_1 loops	$\uparrow$	$\downarrow$	$\cdot$	ringing: less gain
Γ_2 loops	$\uparrow$	$\uparrow$	$\downarrow$	buzzing: less rate
none	$\uparrow$	$\uparrow$	$\uparrow$	all quiet: tighten

Procedure.

1. Close the loop with any gains; conservative ones save iterations.
2. Step the setpoint; record the error e.
3. If the record grows rather than decays (Definition 3), halve all gains and go to step 2.
4. Truncate the record where $|e|$ last leaves 2% of its peak (Definition 2).
5. Form the three portraits (Figs. 2, 4) and count loops.
6. Apply the matching row of Table 1.
7. Repeat.

Table 2: Constants and defaults, used unchanged on every plant in this paper.

Symbol	Meaning	Default
$\bar{\mathbf{N}}$	loop limits	$(0.5, 0.75, 1.0)$
ε	truncation radius	0.1
δ	settling band	0.02
β	step size	0.1
box	multiplier range	$[0.01, 10]$

Formally, the screen verdict takes precedence: an unstable record (Definition 3) halves all three multipliers and no counts are evaluated—the portraits of a diverging error carry no usable information. The backoff is deliberately coarse, $1/2$ rather than $1/\gamma$: instability is a state to be exited quickly, not corrected delicately, and fine adjustment is the business of the count rows once a settling record exists. For a stable record, with the *violated set* and its lowest member

$$V = \{k : N_k > \bar{N}_k\}, \qquad k_{\min} = \min V \tag{6}$$

($k_{\min} = 3$ by convention if $V = \emptyset$), the update reads

$$F^{(k)} \leftarrow \begin{cases} \gamma\, F^{(k)}, & k < k_{\min} \text{ (raise band)}, \\ F^{(k)}/\gamma, & k = k_{\min} \text{ (cut band)}, \\ F^{(k)}, & k > k_{\min} \text{ (untouched)}, \end{cases} \tag{7}$$

for $k_{\min} = 3$ degenerating to raising all bands. Even when $|V| > 1$, only $k_{\min}$ is acted on: by the leakage asymmetry of Section 2 it is the only member of V whose guilt is unambiguous, and bands above it are left alone for lack of evidence, not out of caution.

In loop-shaping terms, the gain profile is pushed up from the low-frequency end, halting and stepping down at the first band that objects: a lexicographic bandwidth maximizer, and a fixed-point search for the boundary of the well-damped set $\mathcal{F} = \{\theta : \mathbf{N}(\theta) \le \bar{\mathbf{N}}\}$.

Three facts follow from (7). Every band the rule pours gain into ($k < k_{\min}$) was verified clean in the current experiment. In logarithmic gain coordinates the corrective moves form a lower-triangular, unimodular matrix, so their compositions reach every point of the multiplicative γ-lattice; an unwanted compensation is cancelled by a lower-band correction one iteration later. And the iteration is an implicit attribution test: violations that clear with the cut at $k_{\min}$ were leakage, violations that persist are genuine and become the new $k_{\min}$; cutting all of V at once would forgo this inference (Section 5).

Remark 3. *Nothing in* (7) *is specific to three channels: any controller whose channels dominate disjoint, ascending frequency bands admits the same construction, with m portraits obtained by walking the derivative chain and the same triangular stencil. It runs downward as readily as up—the nested family* I $\subset$ PI $\subset$ PID *of Assumption 2 is the scheme at* $m = 1, 2, 3$, *so restricting the family restricts the table, not the argument—and fewer bands mean fewer constraints and a lower-dimensional lattice: on P2, from analogous conservative starts, the terminal cycle is entered after* 41 *experiments for PID,* 18 *for PI, and* 6 *for a pure I controller.*

Proposition 6 (Monotone steering)**.** *Under Assumption 1, near the boundary of Proposition 2: (i) along the uniform direction $L \mapsto mL$ every count is strictly increasing in m; (ii) each count N_k is strictly increasing in its own band gain $K^{(k)}$ at a crossover its band owns; the cut of the corrective row of* (7) *strictly decreases $N_{k_{\min}}$, and the compensating raises do not cancel it.*

Proof. Write $\partial N/\partial \log m = (\partial N/\partial \zeta)\,(\partial \zeta/\partial \log m)$. The first factor is negative (Proposition 3). For the second:

the phase-crossover frequencies of mL do not move with m ($\arg mL = \arg L$), while the gain at each grows linearly in m; the margin to the first touch of -1 therefore shrinks strictly, and near the boundary the critical pair's damping follows it (Proposition 2). The product is positive. For (ii), the same computation runs along the band's own gain at the crossover its mode owns (Proposition 4); the compensating raises act on lower bands, subdominant in $|C(j\omega)|$ at that crossover, so one rule step keeps the sign of the cut. □

Remark 4 (Scope)**.** *Proposition 6 is proved near the stability boundary and verified over the decision region by measurement. It is a scalar statement—one count, one loop gain, one crossover—made possible by the single-frequency mechanism of Proposition 2. It does not claim that every pole damping is monotone in every band gain; that matrix-valued statement is false in general (derivative action adds phase lead as well as high-frequency gain) and is never used. Nor is the decrease claimed global in m: strong derivative action can raise the dominant damping over a mid-range of gain—the pair is captured by the controller zeros, which is what derivative action is for—before the fast branches take over; the proof needs only the boundary neighbourhood. On a* 14*-plant FOPTD ensemble with* $\tau = L/(L+T) \in [0.09, 0.90]$*, the raise-all direction showed no above-threshold violations for* $\tau \leq 0.7$*; for delay-dominant plants (*$\tau \gtrsim 0.8$*) the dead-time pole comb crowds the critical pair and isolated count jumps appear, at the* τ *where derivative action is classically marginal [7]. There the iteration stays bounded and parks or cycles.*

Proposition 7 (Behaviour of the iteration)**.** *Unconditionally, the iteration is deterministic and its iterates remain in the multiplier box. Under Assumption 1, via Proposition 6, from any* $\theta^0 \in \mathcal{F}$ *the iteration reaches every neighbourhood of* $\partial\mathcal{F}$ *in finitely many steps—or saturates at the box, if* $\mathcal{F}$ *contains it—and thereafter remains within a band of* $\partial\mathcal{F}$ *of width one multiplicative step per gain, on which it executes a limit cycle.*

Proof. Boundedness is by construction of the projections. On $\mathcal{F}$, the rule ($V = \emptyset$) multiplies all gains by $\gamma > 1$, an expanding map that leaves any compact subset of the interior in finitely many steps or saturates at $F_{\max}$. Once $V \neq \emptyset$, the corrective form of the rule reduces the lowest violated feature by a fixed factor per step (Proposition 6); when it clears, either the leaked violations above it clear with it or a genuine one becomes the new $k_{\min}$ and is reduced in turn, restoring feasibility in finitely many steps; alternation of expansion and correction confines the trajectory to the stated band. □

Remark 5 (Bootstrap from arbitrary gains)**.** *With the screen row, a stabilizing initialization is a convenience, not a precondition: plants of Assumption 1 are open-loop stable, so consecutive halvings reach a stabilizing region within a handful of experiments (or hit* $F_{\min}$*, a detectable outcome), after which Proposition 7 applies unchanged. Each backoff costs one experiment on an unstable loop; the mid-test abort remains the hard safety layer, and a conservative start remains good practice.*

Termination. No stopping rule is needed: the expanding form $V = \emptyset$ is active exactly on $\mathcal{F}$, so the iteration never rests and, by Proposition 7, ends in a limit cycle straddling $\partial\mathcal{F}$ with excursion one step per gain. The limits are set so that the regulators visited by the cycle remain serviceable; the iteration may be left running or capped, keeping whichever iterate it is stopped at. Section 5 quotes the last feasible iterate of each run as a representative of the cycle. The monotonicity that drives the proof is no longer assumed: Proposition 6 derives it from the plant class near the boundary, and Remark 4 delimits its reach. Where the mechanism fails—strong non-minimum-phase behaviour, resonance, or strongly delay-dominant dynamics—the iteration parks at a gain bound: detectable, safe, unproductive. It held on every plant tested. Gains move $\beta = 10\%$ per iteration, and a step test aborts with rollback if $|e|$ exceeds a configured multiple of the step.

Remark 6 (Bumpless implementation)**.** *Because gains change while the loop stays closed, the controller should be realized in* velocity *(incremental) form,*

$$\Delta u_k = K_p(e_k - e_{k-1}) + K_i T_s e_k + \tfrac{K_d}{T_s}(e_k - 2e_{k-1} + e_{k-2}),$$
$$u_k = u_{k-1} + \Delta u_k, \tag{8}$$

rather than positionally: the increment rides on whatever control value is currently applied, so a gain change at an iteration boundary—or a manual-to-automatic transfer—produces no discontinuity in u, and there is no integrator state to reinitialize. This form is assumed throughout Section 5.

5. Validation on a plant battery

The method—guarded features and the rule of Table 1—was run on four process-typical plants spanning lag-dominant to delay-dominant and high-order dynamics:

$$\begin{aligned}
\text{P1 (lag-dominant):}\quad & \tfrac{e^{-s}}{(10s+1)(s+1)^3},\\
\text{P2 (balanced):}\quad & \tfrac{1.25\,e^{-8s}}{(5s+1)^4},\\
\text{P3 (delay-dominant):}\quad & \tfrac{e^{-10s}}{(2s+1)(s+1)^3},\\
\text{P4 (high-order slow):}\quad & \tfrac{e^{-4s}}{(8s+1)^6}.
\end{aligned}$$

All four satisfy Assumption 1: their open-loop step responses are monotone with dead time, even where their order exceeds one (Table 3). The runs use the identical constants $\bar{\mathbf{N}} = (0.5, 0.75, 1.0)$, $\varepsilon = 0.1$, $\delta = 0.02$, $\beta = 0.1$, box $[0.01, 10]$, throughout. In each case the simulated plant serves exclusively as a stand-in for the physical process: it generates step responses; the tuner sees only the recorded error, exactly as on hardware. Initial gains were a nominal design with proportional and integral gains halved and derivative gain tripled—a deliberately mis-tuned start exercising the rule at every band position.

Three observations. First, where the mis-tuned derivative channel creates a fast parasitic mode (P2, P4) the diagnosis is invisible in the raw response but explicit in Γ_2 ($N_2 = 2.7$ for P2, lower bands clean, so the indictment is unambiguous); repeated cuts of band 2 trade D for P and I until the portrait clears, with feasibility first reached at iteration 34. Second, every run ends in the limit cycle of Proposition 7, feasibility revisited within a few iterations of the cap—including P4, which churned indefinitely *without* the guard of Section 3 and converged with it. Third, P1 shows partial saturation: its low band never objects, so F_i rides the box bound while the high band supplies a genuine constraint.

The leakage asymmetry shows live. At iteration 16 of the P2 run, $\mathbf{N} = (0.11, 2.08, 2.22)$: bands 1 and 2 violated with near-equal counts. One cut of band 1 gives (0.11, 0.19, 1.25): the mid violation gone, almost half the high count with it (leakage), and a genuine remainder of 1.25 persisting as the next $k_{\min}$. A variant cutting every violated band at once acts on exactly this unattributable evidence: it drives P2's derivative multiplier to 0.04 and degrades the returned tuning (IAE 10.7 against 6.8 for rule (7)), similarly on P3, and is therefore not adopted despite reaching feasibility faster from severely mis-tuned starts.

The screen row extends the battery beyond stabilizing starts: from eight times the nominal design gains on P2—violently unstable—two halvings settle the response by iteration 2, feasibility follows at 56, and the run returns gains equivalent to (0.86, 1.06, 0.86) times the design (Remark 5).

6. Limits

That a band's oscillation is curable by cutting its gain is derived near the boundary (Proposition 6, Remark 4) rather than presumed; strong non-minimum-phase or resonant dynamics break this, visibly and safely, by parking the iteration at a bound. The sign of N_k is unused; small negative counts (P2 in Table 3) are endpoint artefacts of well-damped records, while a persistent large negative count would signal inverse response and bear on Assumption 1—left to future work. Noise enters the differenced coordinates directly: the guard excludes settled noise under the condition of Remark 2, not broadband noise within the transient, and noisy deployments need filtering outside this paper's scope. Each iteration needs one settled, undisturbed window; a disturbance corrupts one test, not the procedure. Single-band action is the price of attribution: a stable start violating all bands converges through a sequence of single corrections—from $(2K_p, 3K_i, 4K_d)$ on P4, 74 iterations to first feasibility—while the attribution-free uniform backoff is reserved, via the screen, for divergence. Leakage can misdirect a correction when the true offender sits just below its own threshold; the stencil self-corrects within a few iterations, at their cost. The destination is a fast well-damped tuning, not an optimum; where a criterion matters, it is a sound starting point for criterion-based refinement.

7. Conclusion

Three portraits make under-damping visible band by band; a winding count makes it a number; a settling-anchored guard makes the number window-independent; one triangular rule, cutting only the band whose blame is unambiguous, makes it a decision. The features carry no units and no clock, so fixed dimensionless constants serve every plant tested. The iteration is deterministic and bounded unconditionally and converges to the boundary of the well-damped set under a monotonicity derived from the plant class. The method asks one routine setpoint step per iteration, needs no model, and shows its work: an operator watching the three portraits sees every decision the algorithm makes.

Data availability

Reference implementation, simulation code, and scripts reproducing every result: `https://github.com/ottapav/roboPID-simulator`. An interactive demonstration is running at `https://robopid-simulator.onrender.com/`.

Table 3: Plant battery: initial and returned features, returned multipliers, iteration at which the returned (last feasible) tuning occurred, out of 120. Identical constants throughout; rule (7).

Plant	**N** initial	**N** returned	(F_p, F_i, F_d)	iter
P1	$(0.12, 0.13, 0.26)$	$(0.20, 0.11, 0.78)$	$(8.10, 10.00, 4.37)^*$	117
P2	$(0.12, -0.09, 2.69)$	$(0.11, 0.23, 0.00)$	$(1.69, 2.32, 0.25)$	114
P3	$(0.12, 0.09, 0.00)$	$(0.46, 0.68, 0.77)$	$(1.69, 2.87, 0.59)$	112
P4	$(0.12, 0.11, 0.25)$	$(0.12, 0.23, 0.00)$	$(1.37, 2.09, 0.39)$	119

*Integral multiplier at the box bound: the low band never objects for this fast, lag-dominant loop, so F_i saturates at 10, while the high band supplies the binding constraint ($N_2 = 0.78$)—the partial saturation case of Proposition 7.

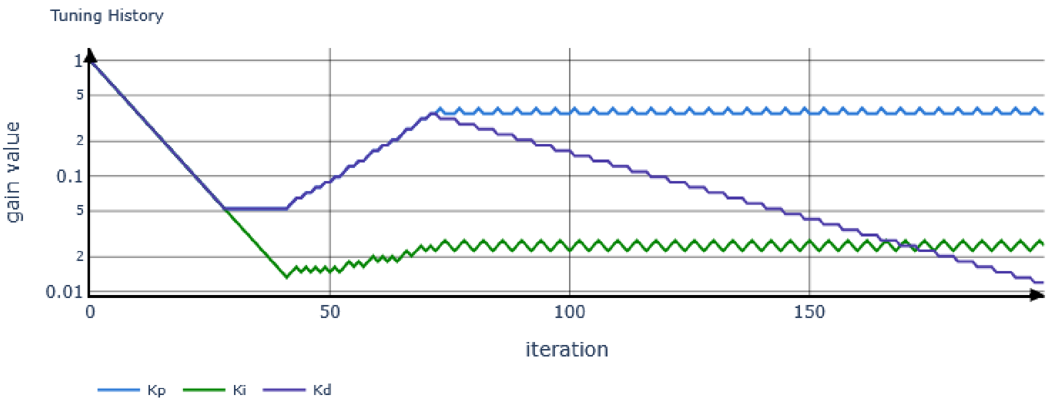

Figure 6: P2, single-band rule: gain trajectories (K_p, K_i, K_d) against iteration, log scale, captured from the running implementation. After an initial corrective phase the gains settle into the boundary limit cycle of Proposition 7.

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